

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Section 1

General

Contents

1.	INTRODUCTION	3
2.	INTRODUCTION TO GYROSCAN INTERA	4
2.1.	Product name, Application and Main features	4
3.	SERVICE ACCESS CONTROL	5
4.	PACKAGES IDENTIFICATION	6
5.	INSTALLATION TIME/SEQUENCE	7
5.1.	Installation sequence	8
5.2.	Installation plan	9
6.	PRE-DELIVERY	15
7.	FACTORY ASSEMBLED MAGNET SYSTEM	16
8.	SITE READINESS CHECKLIST	17
9.	SERVICE TOOLS	20
10.	SAFETY ASPECTS	24
10.1.	General	24
10.2.	Safety with magnetic fields	24
10.3.	Safety with liquid helium or helium gas	25
10.4.	Safety with anaesthetic gases	25
10.5.	Laser light radiation safety	26
10.6.	Electrical safety	28
11.	COOLING SPECIFICATION INTERA STANDARD AND THE OMNI SYSTEMS	29
11.1.	Introduction	29
11.2.	Adjustment (by cooling system supplier)	29
12.	COOLING SYSTEM POWER AND MASTER GRADIENTS	30

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

12.1.	Introduction	30
12.2.	Adjustment (by cooling system supplier)	30

Intera 0.5 T			Intera 1.0 T			Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master			

1. INTRODUCTION

This manual has been written for Philips Medical Systems engineers being involved in installation of a GYROSCAN Intera MR imaging system.

This manual describes the complete INSTALLATION of the Gyroscan Intera system. It describes the **mechanical installation, the setting to work actions** and the **performance measurements**. Section 'Options' shows an overview of options that require installation instructions. The documentation of a number of options is not included in this manual. The documentation for those options is supplied together with the option.

Each section contains a header with a table. This table indicates for what configurations this particular section is valid for. This can be Intera 0.5 T (Standard), Intera 1.0 T (Omni or Power gradients) and Intera 1.5 T (Omni, Power gradients or Master gradients). It is possible that paragraphs within a section are valid for a sub-set of the configurations mentioned in the table. In that case the paragraph heading shows for what configuration this paragraph is valid.

Example:

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni			Omni		

A section that has this table in the header is valid for the Intera 0.5 T (standard), Intera 1.0 T (Omni) and for the Intera 1.5 T (Omni).

For mobile Intera installation a separate "mobile installation" manual is supplied with the system. This mobile manual contains the differences between stationary and mobile systems. The mobile installation manual must be used together with this SMI manual.

Options are no part of the basic installation.

Drawings that are referred to in this manual are part of the DRAWINGS manual Intera.

The installation sequence as shown in paragraph "Installation time" gives the installation steps with the associated Sections. It shows which actions must be finished before starting a new action. Always keep this installation sequence in mind. Deviation from this installation sequence could have a negative effect on the system performance. Throughput time for each action is given.

The site and service requirements mentioned in the Planning Reference Book Intera must be fulfilled, before the installation is started. This to **avoid delays** during the installation.

During the first days of the installation, assistance of a removal crew is necessary for transporting the magnet into the examination room, mounting the gradient coil and transporting all other cabinets into the building.

Arrangements for Helium gas and liquid Helium must be made well in advance with the local supplier. It is the SSD responsibility to make these arrangements.

Arrangements with the supplier of the Hard Copy Unit must be made to install and adjust the Hard Copy Unit. Installation tools must be available on the site. The local organization can buy or rent these tools from the PMG. Some tools have to be arranged locally; see Paragraph "Service tools".

Carefully read the safety instructions in Paragraph "Safety aspects" before starting the installation.

Please, be aware that this medical equipment complies with the requirements of the applicable EMC standards. Electronic equipment exceeding the radiation limits defined in such EMC standards, like GSM and other handheld mobile equipment, may affect the working of our equipment.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
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2. INTRODUCTION TO GYROSCAN INTERA.

2.1. PRODUCT NAME, APPLICATION AND MAIN FEATURES

Philips has been changing how the world looks at MR, since it introduced compact magnets in 1988. The Gyroscan Intera systems bring an entirely new level of MR functionality and efficiency.

- Interactive, real-time imaging, MR's analogy of ultrasound and X-ray fluoroscopy.
- An open environment and in-room, interactive scan control and image (interactive) display that let you stay in touch with your patient and the exam.
- Image reconstruction speed (Compaq XP1000 reconstructor) to keep ahead of the video-like display as you navigate through the anatomy.
- The power and flexibility to handle any exam, from neuro to cardiac to spectroscopy to applications that push the envelope.
- Workflow enhancements that minimize cost and time.
- An advanced technology platform that keeps your MR investment future-safe.
- Mobile systems that travel on field, saving time and money.
- The wide range of RF coils produce state-of-the-art results in all anatomic areas.

The Gyroscan Intera family shares a proven concept of compact magnet technology, low cost of ownership and sophisticated, user-friendly design and seamless integration in the hospital infrastructure.

- Gyroscan Intera 0.5T
 - Standard Gradients (Copley 234, air-cooled gradient coil).
- Gyroscan Intera 1.0T
 - Omni Gradients (Copley 234/P04, air-cooled gradient coil).
 - Power Gradients (1* Copley 274, water-cooled gradient coil).
- Gyroscan Intera 1.5T
 - Omni Gradients.
 - Power Gradients.
 - Master Gradients (2* Copley 274, water-cooled gradient coil).
- Gyroscan Intera CV
 - Gyroscan Intera 1.5 T, Master Gradients, Including all cardiac and vascular options.
- Gyroscan Intera I/T
 - Gyroscan Intera 1.5 T, Master Gradients, Including all interactive and therapy options.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

3. SERVICE ACCESS CONTROL

From R6.1.2 onwards, access to the service functions in the software by unauthorized persons is avoided by using a so-called challenge and response mechanism. This means that before protected service functions can be accessed, the engineer will have to supply the correct response to the challenge (string of 8 characters) generated by the system. **All service functions needed during the installation can be accessed without the response mechanism.**

More information about this procedure can be found in newsletter MR105 "service access". It must be realized that a PC program, which uses the hardware key (dongle), generates this response. The response generator software can be downloaded from the BBS (section MR download area 13 NT Systems) and can be found on the customer support CD-ROM.

An authorization string to update the hardware key can be ordered using the same procedure as used for DSS and other (hardware key resident) authorizations. More information about the implementation of the service access mechanism in the Gyroscan software can be found in chapter 21.2 "service access" of this manual.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

4. PACKAGES IDENTIFICATION

A Gyroscan Intera system is delivered in about 30 packages (also called crates or collies). On the packing-list a package is called: "COLLI FOLLOW NR."

Each package is marked with a sticker with a six-digit number. The first four digits are used for system (site) identification and is equal for ALL (!) packages of that system. In this way an easy check can be done that no exchange of packages has been taken place. The last two digits are the actual package numbers.

Example: number 325901 => system/site 3259, package 01.

Figure 1 - PACKAGE Identification

3259 01

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

5. INSTALLATION TIME/SEQUENCE

The table on the next page shows a normal installation sequence. Use this table as a guide to go through the sections of this manual.

For this year 2000, the standard installation time was set to:

Site assembled magnet systems:

Installation labor : 200 man hours
Installation day's : 12

Factory assembled magnet systems:

Installation labor : 160 man hours
Installation day's : 10

Extra time is needed for major options as shown in the table.

These mayor options require extra time	Nr. Eng	man hours
Power Gradients	1	20
Master Gradients	1	25
Easy vision MR	1	8
Synergy	1	8
Multi nuclei	1	11
Proton Spectroscopy	1	4
Hospital network connectivity	1	6

The mentioned installation time above can only be realized if pre-conditions are fulfilled.
Site management must be well organized by local organization, e.g:

- Magnet arrived on LHe temperature.
- Site prepared (see paragraph "Site Readiness Checklist" (see also Planning Reference Book).
- A removal crew with equipment is available during the first days of the mechanical installation.
- Cryogenics for refilling the magnet.
- Tools ordered locally or from PMSN (see paragraph "Service tools").
- Hard Copy Unit representative to check unit at time of performance measurements.
- A pre-installation kit must be installed prior to this installation (refer to next paragraph).
- A mechanical installation crew for the first days can decrease the installation throughput time.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

5.1. INSTALLATION SEQUENCE

The Installation sequence is displayed in the next chart. It explains how to go through the sections of this manual. Note (see chart) that you will not go through the sections in a sequential order.

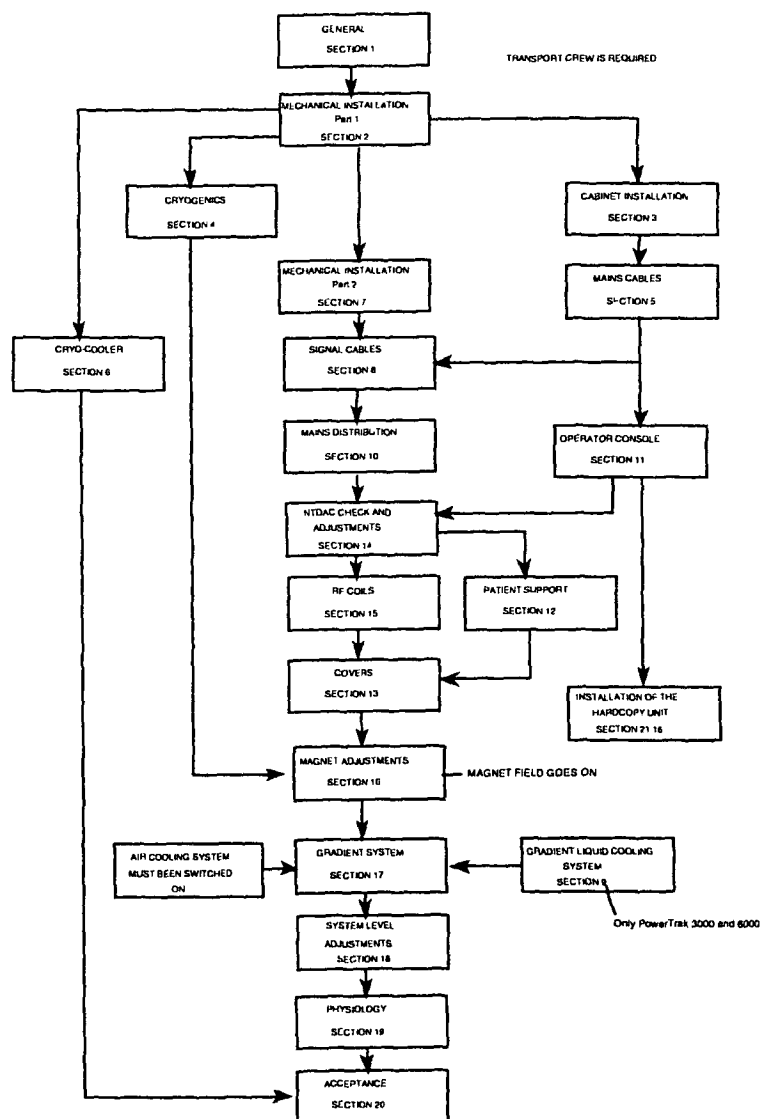


Figure 4 - Installation sequence

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

5.2. INSTALLATION PLAN

This paragraph shows an example of a standard Intera "day by day" installation plan. In this installation plan, the pre-installation kit is planned to be installed in advance by RF cage supplier and local hospital electrician. The plan contains a factory assembled magnet system.

The plan shown does not include delays such as:

- Construction delay
- Missing parts, tools etc.

ACTIVITIES DAY 1	SMI section	Nr. Eng.	man hours	others
Unpacking/transport into hospital	6	1	6	crew
Position and align magnet	2	2	4	crew
Install gradient coil (+ ring frames) *2)	2	(1)	(3)	(crew)
Install MDU *1)	3			Loc. electrician
Install S.F.B	3	1	0.5	crew
Position NTDAC	3	1	0.1	crew
Position power amplifier	3	1	0.1	crew
Position Gradient amplifier	3	1	0.1	crew
Position refrigerator system compressor	3	1	0.1	crew
Mount cryo-cooler transformer (not for USA) *1)	3			Loc. electrician
Mount patient ventilation unit	3	2	1	
Position Operators console	3	1	0.5	crew
Install Quench buttons	3	1	1	
Install mains stabilizer (USA ONLY)	3	1	0.5	Loc. electrician
Install MR5002 into NTDAC	3	1	0.5	crew

*1) Must be installed previously (Pre-delivery kit)

*2) Already factory mounted if you have a factory assembled magnet system

Manpower:

- Transport crew with equipment must be available.
- 2 engineers needed

Tools:

- Electric screwdriver
- Standard toolbox
- Spirit level hose (transparent)
- Crowbar
- Pallet truck with forks > 150 cm
- Magnet transport set T5/NT/Intera
- Hex Screw driver 5x300 mm for gradient coil installation
- Electrical drill (adjustable torque + bits)

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
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ACTIVITIES DAY 2	SMI section	Nr. Eng.	man hours	others
Helium level check	4	1	0.1	
LHe filling if needed (average)	4	1	1.5	Gas-company
Install gas exhaust interface *1)	2			
Install gas exhaust line	2	1	2	
Install PFEI *2)	2	(1)	(1)	
Install Coverframe *2)	2	(1)	(1)	
Mount air guiding shields (only in standard and Omni configuration)	2	2	4	
Mount air ducts *2)	2	(2)	(2)	
Mount patient cooling duct		1	1	
Mount external air hoses	2	1	0.5	
Mount PICU	2	1	0.25	
Install gas-line interface plate *1)	6			

*1) Must be installed previously (Pre-delivery kit).

*2) Already factory mounted if you have a factory assembled magnet system

Manpower:

- Gas-company for :Helium filling
- Plumber to connect water to refrigerator compressor and heat-exchanger
- 2 engineers needed

Tools:

- He Transfer line (if LHe filling is needed)
- Gloves and face shield for cryogenic work
- Heat gun

ACTIVITIES DAY 3	SMI section	Nr. Eng.	man hours	others
Install gas line	6	2	6	
Connect compressor to water	6	1	0.5	
mains, earth and signal cables	5+8	3	12 (36)	

*1) Must be installed previously (Pre-delivery kit)

Manpower:

- Three engineers
- Extra manpower can speed this action

Tools:

- Standard tool box
- 2 step ladders
- Cable pull 'snake'
- Electrical jigsaw + blades for metal
- AMP Clamp tool (Power and Master gradients only)
- Torque wrench 30-140 Nm (Power and Master gradients only)
- Tube bender 17 mm 5/8" for gas lines

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

ACTIVITIES DAY 4	SMI section	Nr. Eng.	man hours	others
Setting to work the computer	11	1	0.5	
Configuring software	11	1	0.5	
Test intercom	11	1	0.1	
Insulation measurements	10	1	0.5	
Earth bounding tests	10	1	1.5	
Check mains connections	10	1	0.1	
Modify RF amplifier to 60 Hz (USA only)	10	1	0.5	
Check/Adjust Body coil	15	1	1.75	
Modify head coil for right side PICU	15	1	0.5	
Check/Adjust head coil	15	1	1.75	
Level and align patient support	12	2	2	
Setting to work the patient support	12	1	0.5	

Manpower:

- 2 engineers

Tools:

- Earth bounding tester
- Multimeter

ACTIVITIES DAY 5	SMI section	Nr. Eng.	man hours	others
Install the magnet covers	13	2	6	
Install the Patient support covers	13	1	1	
Install the covers of the system filter box	13	1	0.5	
First time power up (NTDAC)	14	1	0.1	
Parameter settings	14	1	0.5	
NTDAC tests and adjustments	14	1	1.5	
Setting to work the magnet (ramp up)	16	1	2	

Manpower:

- 2 engineers

Tools:

- Electrical jig saw
- Magnet power supply tool
- Ramp down accelerator
- Shim Tesla meter (METROLAB)
- Shim gantry NT/Intera
- Foot-switch for shimming

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

ACTIVITIES DAY 6 + 7	SMI section	Nr. Eng.	man hours	others
Shimming the magnet	16	2	20	
Final actions after shimming	16	1	4	

Manpower:

- 2 engineers

Tools:

- Magnet power supply tool
- Ramp down accelerator
- Shim Tesla meter (METROLAB)
- Shim gantry NT
- Foot-switch for shimming

ACTIVITIES DAY 7	SMI section	Nr. Eng.	man hours	others
Setting to work the gradient system	17	1	1	
Eddy current adjustments	17	1	4	

Manpower:

- 1 engineer

Tools:

- Eddy current compensation set
- Dual mode adjustment tool (Master gradients only)

ACTIVITIES DAY 8	SMI section	Nr. Eng.	man hours	others
System Res. Freq. Adjustments	18	1	0.5	
RF power ref. and PU coil adj.	18	1	0.5	
Gradient system	18	1	1	
Total backup	18	1	0.25	
MRL system acceptance	18	1	0.5	
Safety aspect tests	18	1	0.25	
PATINT tests	19	1	0.5	
PICU physiology automatic test	19	1	0.5	
Nurse call	19	1	0.1	
Artifact measurements	20	1	3.5	
PIQT	20	1	0.5	

Manpower:

- 1 engineer needed

Tools:

- Phantom set
- Current clamp (HIOKI 3263-01)
- non magnetic tool set

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

ACTIVITIES DAY 9	SMI section	Nr. Eng.	man hours	others
Head coil	20	1	2	
Body coils	20	1	1.75	
Standard supplied surface coils	20	1	0.5	
Hardcopy unit	21.4	1	3	
Other Surface coils avg 3 per system	21.13	1	0.75	

Manpower:

- 1 engineer needed

Tools:

- Phantom set
- Test box Hard Copy Unit
- Sensito-meter
- Densito-meter

ACTIVITIES DAY 10	SMI section	Nr. Eng.	man hours	others
Other minor options (average)	21.1	1	4	
Unforeseen	-	1	2	
Cleaning (average)	-	1	2	

Manpower:

- 1 engineer needed
- Arrange manpower to send back all tools.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Other minor options as planned for day 10 are included in the installation time as mentioned earlier. The next table shows these minor options.

Other minor options are:	Nr. Eng.	man hours
Trolley	1	3
Hard disk	1	0.75
5,25" cartridge set 1.3 Gb	1	1
patient Observation System	1	4
Spectroscopy (1H (1.5T only)	1	4
Audio	1	2
Reconstructor 512 Mbyte option	1	2
Front End Display	1	3
Video option	1	2

The following major options require extra installation time. Both man hours and throughput time will increase. Hours for these options are booked separately in the local administration systems.

These major options require extra	Nr. Eng.	man hours
Power gradients	1	20
Master gradients	1	25
Easy vision MR	1	8
Synergy	1	8
Multi nuclei	1	11
Proton Spectroscopy	1	4
Hospital network connectivity	1	6

NOTE

Be aware that delays are not included in the above mentioned installation time. Hours spend at other activities must **not** be booked as "system installation hours". These activities can be:

- Repair actions
- Missing items or tools
- damage in shipment
- Construction delay
- planning errors

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

6. PRE-DELIVERY

It is possible to order parts of the equipment in advanced. These system parts than can be installed prior to the actual installation.

A special "Pre-installation parts" MMR0241 is available. On the order, the Required Delivery Date (RDD) can be set to an earlier date than the rest of the system delivery.

Contents of MMR0241

P/N	Description	Contents
9896 030 04632	Refnegerator system transformer 6KW	not for USA
9896 030 02551	MDU extended	Only Power and Master gradients
	Noise reduction disks	All configurations
	Magnet positioning jig	All configurations
9896 030 02561	pre-installation parts 4522 131 54961 Mains Distribution Unit 4522 130 65982 Helium exhaust feedthrough 4522 131 54941 Mount. mat. He exhaust 4522 131 57082 Hel gas line RF feedthrough 4522 131 54952 Mount. mat. He. gas line feedthrough	All configurations

- Advantages:
- MDU and MDU extended can be mounted to the wall. The hospital mains can be connected by local electrical
 - The Helium exhaust RF feedthrough can be mounted by the RF cage supplier instead of a dummy plate. The external exhaust system can be connected.
 - The Helium gas lines feedthrough can be mounted by the RF cage supplier instead of a dummy plate.

By doing so, the actual installation time can be reduced with approximately 8 man hours.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

7. FACTORY ASSEMBLED MAGNET SYSTEM

In order to reduce the installation time with 40 man hours, it is possible to order a factory assembled magnet system. Obviously, factory assembly can only be carried out for magnet that are shipped from Best. For magnets transported directly from the magnet manufacturer to the customer's site ("drop-shipments"), the existing situation remains unchanged. This is the case for systems with destinations in the USA and Japan.

In order to provide the SSD's with a choice between factory assembled magnet systems and the current way of delivery, a questionnaire will be included in the catalogue. **The default answer will be "factory assembled magnet system: yes".**

The SSD logistics department can request changing from "factory assembly: yes" to "factory assembly: no", or vice versa, until 6 weeks before system shipment.

In a factory assembled magnet system, a large number of parts and assemblies are mounted at the factory in Best. These include:

- Gradient- /QBC combination
- Gradient carriers
- Ring frame rear and sub-frames front
- PFEI
- Air ducts
- Bridge section
- Most of the magnet cabling including gradient cabling.
- gradient strips (Master gradients and Power gradients only)

Note that section 2 is divided into two sections:

- First section contains procedures for factory assembled magnet systems.
- Second section contains procedures for non factory assembled magnet systems

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

8. SITE READINESS CHECKLIST

Prior to the installation the Site Readiness must be checked using the Site Readiness checklist and the RF shielded room test certificate both available in the Planned Reference Book. Only use the next checklist if the Site Readiness was not checked previously. The site readiness will have a strong influence on the installation time.

Not all check points will be applicable, this depends on local regulations and site circumstances.

HOSPITAL :

SRN :

1 – General

Appointment made with supplier of hard copy unit to install and adjust the HCU during installation	YES/NO
Liquid helium available at start of installation	YES/NO/N.A.
Gradient cooling system complete according specification	YES/NO
Air conditioning working in all rooms	YES/NO
All rooms finished and cleaned	YES/NO
All Service tools are available (see paragraph "Service Tools").	YES/NO

2 - Transport

Magnet transfer opening in RF room checked	YES/NO
Transport route for maximum floor loads checked	YES/NO
Route for minimal dimensions checked	YES/NO
Route and unloading area cleared from obstacles	YES/NO
Size of dewar to pass through building and examination room door	YES/NO

3 - Examination room

RF shielding certified	YES/NO
RF enclosure / RF door / RF window	YES/NO
Magnet transport opening checked for min. dimensions	YES/NO
Suspended cable tray to magnet present	YES/NO
Protective earth terminal present	YES/NO

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Air inlet/outlet openings (wave-guides) present	YES/NO
Mounting pads magnet (if applicable) and table located correctly	YES/NO
Cable duct to patient support located correctly	YES/NO
Penetration panel helium gas lines present	YES/NO
Penetration panel helium exhaust pipe present	YES/NO
Penetration panel future use present (See Planning Reference Book)	YES/NO
AC Auxiliary Power filters and signal filters present	YES/NO
Door switch present	YES/NO
Fixing point ERDU button defined	YES/NO
Fixing point Patient observation camera('s) present	YES/NO
Emergency lighting present	YES/NO
Suspended ceiling present	YES/NO
Internal wall finishing complete	YES/NO
External wall finishing complete	YES/NO
Floor finishing complete	YES/NO
Airco pipes above suspended ceiling present	YES/NO
Air diffusers in suspended ceiling present	YES/NO
Auxiliary wall sockets	YES/NO
Service wall sockets above magnet present	YES/NO
Sound absorbing material for walls and suspended ceiling present	YES/NO/N.A.
Provisions for medical gasses present	YES/NO/N.A.
Oxygen level detector present	YES/NO/N.A.
Removable suspended ceiling at magnet present	YES/NO
Grounding stud to connect potential equalizing busbar	YES/NO
Earth point in RF cage near to System Filter Box present	YES/NO
Fixation points speaker (2x) present	YES/NO
Fixation Interactive display	YES/NO/N.A.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

4 - Control room

Sound absorbing material in suspended ceiling present	YES/NO/N.A.
Cable duct to technical room present	YES/NO
Telephone lines (2x) installed	YES/NO
Modem for remote service available	YES/NO

5 - Technical room

Mains according specifications (see Planning Reference Book)	YES/NO
Mains impedance according specification	YES/NO
Mains, switch and fuses connected	YES/NO
Arrangements made for cable connection to MDU	YES/NO
Switch and fuses for cryocooler present	YES/NO/N.A.
Switch and fuses for chiller present	YES/NO/N.A.
MDU mounting studs present	YES/NO
Cryocooler transformer mounting studs present	YES/NO/N.A.
Water supply and return installed for cryocooler	YES/NO
Helium gas supply system installed	YES/NO
MDU Extended studs present	YES/NO/N.A.
Patient ventilation studs present	YES/NO

N.A = Not Applicable

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

9. SERVICE TOOLS

A number of special tools are needed during:

- Pre-Installation
- Transport and unpacking
- Installation, corrective and planned maintenance

Some of the tools needed are very expensive or seldom used. Philips Medical Systems - Best stocks a number of these tools which can be rented through the PMG spare parts department.

(Pr) Pre-Installation tools (All Intera configurations)

The following tools are used for GYROSCAN systems.

Item	Note	Type	Name/12NC	Supply
Mains resistance meter (Panensa MIC-5)	*1)	Pr	4522 980 35231	PMSN
Mains analyzer (BMI 4800 power scope)	*1)	Pr	4522 980 37421	PMSN
B0 measuring set	*1)	Pr	8122 101 89691	PMSN
RF measuring set	*1)	Pr	4522 980 39571	PMSN

Rent/WK:

(FL..) = Tool can be rented

(-) = Tool can only be bought

*1) Refer to the Planning Reference Book for information on when to use these tools.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

(I) Installation, (C) Corrective maintenance and (P) Planned Maintenance tools

Tools are use with GYROSCAN systems.

Item	Note	Type	Name/12NC	Supply
Earth bounding tester		I/C/P	4522 980 38072	PMSN
Multimeter (Fluke 87)		I/C/P	4523 500 58221	PMSN
Gloves and face shield for cryogenic work		I/C/P		Local
He Transfer line	*8)	I/C/P	(see note *8)	PMSN
Standard tool kit		I/C/P	4522 980 38282	PMSN
Mains extension cable		I/C/P	15 m / 3 socket	Local
Solder station		I/C/P		Local
Heat gun		I/C/P	1500 watt	Local
Torch		I/C/P		Local
Test box Hard Copy Unit		I/C/P	8122 102 22601	PMSN
Service P.C	*1)	I/C/P		Local
Current clamp (HIOKI 3263-01)		I/C/P	4522 980 37661	PMSN
None magnetic toolset	*9)	I/P/C	4522 130 54415	PMSN
Magnet Power supply tool		I/P/C	4522 150 15801	PMSN
Ramp down accelerator		I/P/C	4522 150 15811	PMSN
Eddy current comp set	*4)	I/C	8122 101 45004	PMSN
Dual mode adj tool (only for Master gradients)		I/C	8122 102 64411	PMSN
Shim Tesla meter (METROLAB)		I/C	4522 980 35272	PMSN
Shim gantry NT/Intera	*5)	I/C	8122 102 20356	PMSN
Torque wrench 30-140 Nm		I/C	2622 150 20193	PMSN
Hex Screw driver 5x300 mm for gradient coil installation	*8)	I	8122 102 54731	PMSN
Electrcal jigsaw + blades for metal		I		Local
Electric drill (adjustable torque)		I	Left/Right	Local
Screw driver bits for drill		I		Local
Sanding disk for electric drill		I		Local
Step Ladder		I		Local
Plummet		I		Local
Spirit level hose (transparent)		I	5 meter	Local
Spirit level		I	50 cm - 0.5 mm/m	Local
Crowbar		I	2 m	Local
Cable pull 'snake'		I		Local
Pallet truck with forks > 150 cm		I	1000 kg	Local
Magnet transport set T5/NT	*6)	I	8122 101 76403	PMSN
Tube bender 17 mm 5/8" for gas lines		I	8122 102 59951	PMSN
AMP clamp tool (Only Power and Master gradients)		I	8122 102 67551	PMSN
Magnet rotation tool	*10)	I	9896 030 01751	PMSN
Gauss meter + probe		C	4522 980 35261	PMSN
Portable He leak tester		C	4522 980 39441	PMSN
Manometer for cooling system		C	0-200 [Pa]	Local
ECG simulator		C	4522 980 37681	PMSN
Decision Support System	*2)	C		PMSN
Dicom Protocol Monitor (see Newsletter ICS-011)		C	4522 500 66021	PMSN
Dicom Protocol Monitor + NetXray		C	4522 500 66011	PMSN
Printer test tool for H.C.U		C	4522 980 38561	PMSN

Rent/WK: (FL..) = Tool can be rented
 (-) = Tool can only be bought

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

*1)	Specifications Service PC Minimum. 386 / 5 Mb int. mem. / 80 Mb Hard Disk / Mouse ore track ball. Recommended: 486DX / 8 Mb int. mem. / 200 Mb Hard Disk / Mouse ore track ball / pcmcia slots (2x type II)																								
*2)	Decision Support System (including electronic documentation) can be obtained from PMSN (refer to tool catalog for order information).																								
*3)	Coldhead extractor tool only needed if ceiling height is below 2 90 meters																								
*4)	The E.C. comp set 8122 101 45003 must be upgraded to a E.C. comp set 8122 101 45004 Procedure Please contact Return Shipment Department, Service logistics in Best for a Return Authorization number (RA number) for customized repair according to usual procedure. Refer to Procedure for updating E.C comp set to "8122 101 45004 or higher" We expect you to return code-number 8122 101 4500(<4) and after update we will return the same under new 12 nc 8122 101 45004 or higher. To monitor the Eddy current curves the following is possible: 1) OCD monitor 2) Local (S)-VGA computer monitor Parts which can be ordered: <table><tr><th>Description</th><th>Length</th><th>Option</th><th>Codenummer</th></tr><tr><td>BNC-BNC video cable (3 wires)</td><td>30 m</td><td>1</td><td>4522 131 32514</td></tr><tr><td>Keyboard ext. cable</td><td>6 m/piece</td><td>1 / 2</td><td>4522 131 50671</td></tr><tr><td>Mouse ext. cable</td><td>6 m/piece</td><td>1 / 2</td><td>4522 131 50671</td></tr><tr><td>VGA-VGA ext. cable</td><td>10 m/piece</td><td>2</td><td>4522 150 20241</td></tr></table> <i>Note 1:</i> The number of extension pieces of the keyboard, mouse and VGA-VGA cables can be different for every site, so you have to order the number of extension cables that you need. <i>Note 2:</i> If you choose option 1 you will need to order 2x BNC-BNC video cables because of the 2 extra sync signals.					Description	Length	Option	Codenummer	BNC-BNC video cable (3 wires)	30 m	1	4522 131 32514	Keyboard ext. cable	6 m/piece	1 / 2	4522 131 50671	Mouse ext. cable	6 m/piece	1 / 2	4522 131 50671	VGA-VGA ext. cable	10 m/piece	2	4522 150 20241
Description	Length	Option	Codenummer																						
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Mouse ext. cable	6 m/piece	1 / 2	4522 131 50671																						
VGA-VGA ext. cable	10 m/piece	2	4522 150 20241																						
*5)	8122 102 20356 Shim gantry + foot-switch + 4 bolts + socket 11 mm + shimgantry interface The Shim gantry T5 12NC 8122 101 80001 can be upgraded to a "universal shim-gantry NT/T5/Intera" Procedure: Please contact Return Shipment Department, Service logistics in Best for a Return Authorization number (RA number) for customized repair according to usual procedure. Refer to: Procedure for updating T5 shim-gantry to "universal shim gantry T5/NT/Intera". We expect you to return codenummer 8122 101 80001 and after update we will return the same under new 12 nc 8122 102 23066 as "Universal shim gantry T5/NT/Intera" The "Universal shim gantry T5/NT/Intera" cannot be rented.																								
*6)	The "Magnet transport set T5/NT" 12NC 8122 101 76403 can be updated to Magnet transport set T5/NT/Intera with "transportset update kit" 8122 102 72981.																								
*7)	Philips 12 nc	Vendor p/n	Vendor	Min ceiling height NT	Min ceiling height F2000																				
	4522 300 02892	621.0635	OMT	2716	3080																				
	4522 150 00142	20321	IGC-T5	2736	3186																				
	4522 150 00143	30114-01	IGC-T5/NT	2620	3186																				
	4522 150 23101	n.a	Air products	2555	2885																				
	4522 150 23101 + 23481	n.a.	Air products	--	2650																				
	Not available	On request	Linde L-form	2565	3085																				
	Not available	On request	Linde -dip tube	Inserting outside																					
	Not available	3 221 3451	Linde U-form	2735	2915																				
	4522 150 23611	n.a.	IGC-F2000	2620	2620																				
*8)	The "Hex screwdriver 5x300" 8122 102 54731 is also part of the "shim gantry NT" 8122 102 20351																								

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

*9)	The non-magnetic special tool box is not supplied with the equipment. This non-magnetic tool box must be part of the MR Service Engineer tool box 1x Screwdriver - 3/32x3" 1x Screwdriver - 3/16x6" 1x Screwdriver - 1/4x5" 1x Hex key wrench - 3mm 1x Hex key wrench - 4mm 1x Hex key wrench - 5mm 1x Combination wrench - 10 mm 1x Combination wrench - 13 mm 1x Combination wrench - 17 mm 1x Combination wrench - 19 mm 1x Open end wrench - 5x5.5 mm 1x Open end wrench - 6x7 mm 1x Open end wrench - 8x9 mm 1x Needle nose Plier 1x Tweezers 1x Folding rule - 2m
*10)	New "half moon" pieces are needed for the F2000 B/C rotation tool. Distribution via the "Commercial logistics" department

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

10. SAFETY ASPECTS

10.1. GENERAL

The Gyroscan system is designed and manufactured according to the International Standards for Medical Equipment. Be cautious when accessing, handling, testing, adjusting or servicing the system. Carefully study the following paragraphs.

10.2. SAFETY WITH MAGNETIC FIELDS

It is the responsibility of the hospital that the following safety requirements are met.

- During the site planning of the Philips Gyroscan system, a controlled zone must be defined where the field strength is more than 0.5 mT (= 5 Gauss) has to be defined. Warning signs "CAUTION - Magnetic field permanently switched on" should be used to indicate this area.
- Persons having a pacemaker, neurostimulator, insulin pump or other biostimulation device, implants consisting of ferro-magnetic material, such as surgical clips, artificial cardiac valves and prosthesis or metal splinters, must not be brought into a strong magnetic field. Such persons must stay outside the "controlled zone".
- Ferro-magnetic objects, such as scissors, tools, gas-bottles, vacuum cleaners and iron stretchers, must not be brought into the neighborhood of the magnet and must therefore be kept outside the examination room. Such objects will be attracted by the magnet and may cause injury to the service personnel, patients, staff and/or damage to the equipment.
- Information on magnetic data carriers such as floppy disks, tapes and credit cards can be erased by the magnetic field. A safe limit is 1 mT (= 10 Gauss).
- The security procedures at the entrances of the examination room should prevent that ferro magnetic objects being brought into the examination room. Metal detection equipment can be used. During installation place the safety warnings available in the drawings manual Z9-2.1 and Z9-2.2 to the main entrance of the examination room before energizing the magnet. After the installation the safety drawings must be replaced by proper warnings (responsibility of customer)
- The Gyroscan system is provided with a magnet emergency rundown unit with two remote push buttons for shutting down the magnetic field, which should only be used in case of emergency.
- The patient must be removed from the examination room should medical instruments be needed in a medical emergency.
- Under no circumstances should a container with liquid helium be brought into the magnet area unless it is known that the container is made of non-ferrous material. Special non-ferrous containers are available from the liquid gas suppliers and must be appropriately labeled as non-ferrous containers.
- In case of a deliberate quench by the operator, to implement adequate life supporting and other safety procedures, the magnet field strength at the iso-centre is reduced to a value smaller than 20 mT within 10 seconds.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

10.3. SAFETY WITH LIQUID HELIUM OR HELIUM GAS

Helium gas is odorless, non-flammable and non poisonous. During evaporation, a cold haze is formed. Helium gas will rise to the ceiling. The gas is harmful to health as it will dilute the oxygen in the air.

- Liquid helium is extremely cold and will cause frost bite when in contact with the human body. Use protective gloves, goggles and clothing when handling liquid helium.
- Cryogenic liquids should be handled only by properly-trained staff.
- Always keep the ventilation of the examination room running.
- The magnet system has to be connected to a dedicated venting system, the helium gas exhaust pipe, which prevents evaporation of helium into the examination room.
- The helium gas exhaust pipe opening should be located at a non accessible area. In case of a quench, the helium gas should not affect the oxygen concentration in the accessible area.
- Under no circumstances should the magnet be energized prior to the installation of the helium gas exhaust pipe and the emergency run-down unit.
- Monitoring of the oxygen content of the ambient air, may be obligatory by local regulations. During refill of liquid helium an amount of helium gas will evaporate in the examination room and will dilute the oxygen in the air. For sites with small examinations rooms or low ceilings, it is highly recommended to install an oxygen detector with audible alarm, and a remote sensor on top of the magnet. In case of low oxygen level the engineer will be warned.

10.4. SAFETY WITH ANAESTHETIC GASES

The Gyroscan system is not certified as Anaesthetic Proof equipment.
Non flammable gases can be used.

NOTE

- *Some disinfectants vaporize to form explosive mixtures. If such products are used the vapor must be allowed to disperse before the equipment is returned to use.*
 - *Provisions for medical gases (if required/permitted) preferably mounted at the patient support side of the magnet.*
-

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

10.5. LASER LIGHT RADIATION SAFETY

The laser light is used for orthopaedic alignment and is fitted on the front cone cover. The laser produces a laser light cross-beam, which is visible red and projected directly through an aperture, at the top front cone cover, vertically downwards.

The laser illuminates if the button  on the front PICU is activated. After a period of 15 seconds the laser will shut off automatically.

The laser product consists of an FDA CLASS II, IEC Class 2 Laser.

Laser illumination duration is 15 seconds, wavelength is 635 nm and the power is 0.35 mWatt.

The laser light in the laser alignment tool should only be used under supervision of a medical trained person with knowledge of the hazards implied by the use of a laser light.

It is the user's responsibility to fulfil the local safety regulations regarding laser light radiation.

WARNING



- Class II (FDA) / Class 2 (IEC) laser, do not stare into beam.
- The laser must not be switched on without purpose, and unnecessary exposure must be avoided.
- Use of controls, adjustments or procedures other than those specified in this manual may result in hazardous radiation exposure.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

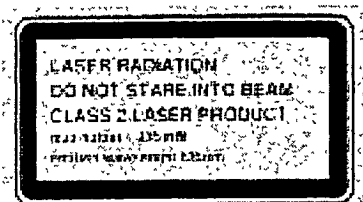
Labelling:

The MRI scanner's alignment tool has the following labels:

At the laser aperture (outside front cover):

Laser aperture

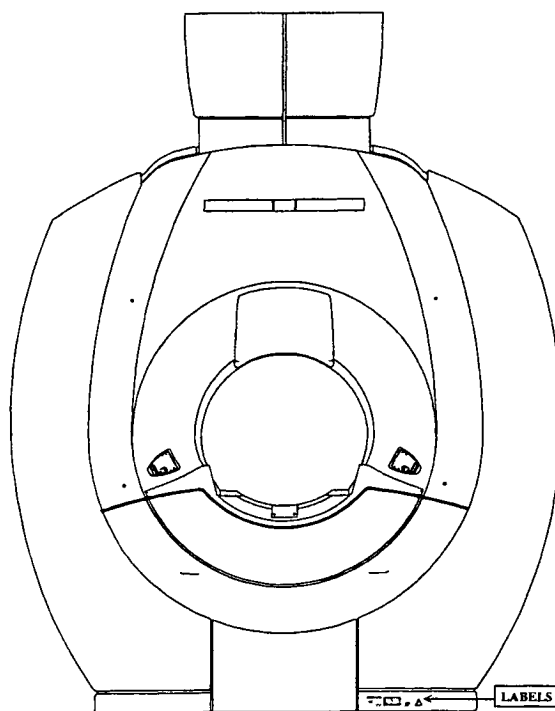
Close to the system identification (outside back cover):



CAUTION
LASER RADIATION
MAX OUTPUT < 0.35 mW
EMITTED WAVELENGTH 635nm
CLASS II LASER PRODUCT

LASER RADIATION
DO NOT STARE INTO BEAM
CLASS 2 LASER PRODUCT
MAX OUTPUT < 0.35 mW
EMITTED WAVELENGTH 635 nm

Figure 2 – Location of labels



Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

10.6. ELECTRICAL SAFETY

- Always be sure that the power is switched off when mounting parts to equipment. If large internal capacitance have been used, be sure that they are completely discharged. Discharging can take several minutes
- Do not measure or adjust equipment with power on unless another person is present.
- Do not allow water or other liquids to enter the equipment, as they may cause short-circuits or corrosion.

- **Gradient system**

Be aware of the high voltage (420 Volt) present inside the gradient amplifier cabinet(s), on the connections to the gradient coil and the system filter box.

Make sure that the AC power has been switched off for at least 2 minutes before starting to work on the gradient chain. It is advised to watch the high voltage value on the EMI 20kW/420V DC power supply meters and make sure that the supply capacitors have discharge to less than 2 Volt.

The cabinet is designed to protect operating personnel from internal hazardous voltages. Operation with unlocked or missing doors should be attempted only by qualified service personnel.

Do not remove ventilation units in front of the power modules, when the gradient cabinet is switched on.

Wear safety glasses during servicing to prevent eye injury from flying debris in the event of an arc.

Only one 20 kW/420V EMI power supply should be extended on the extension rails at a time. If more than one power supply is extended, the rack could tip over.

- **RF power amplifier**

The RF amplifiers must be installed, operated, and serviced by qualified personnel.

Depending on the amplifier model, the RF amplifiers are capable of delivering up to 25000 Watts peak (1250 watts average) of power, into a 50 ohm load. These levels of RF power can cause severe RF burns and other injuries if misused. Never operate an amplifier without having the RF output terminated into the load of the MR system or having the RF input disconnected.

Hazardous voltage can exist at the output of these power amplifiers. Therefore it is advised to disconnect the RF and gating input of the amplifiers before disconnecting the RF output cable.

Lethal voltages are present inside the power amplifiers. Do not operate the amplifiers with covers removed. Consult the Technical manual supplied with the amplifier before servicing.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

11. COOLING SPECIFICATION INTERA STANDARD AND THE OMNI SYSTEMS

11.1. INTRODUCTION

The gradient coil must be cooled by an air-flow of 800 - 1000 m³/h. This amount of air must be sucked out of the MR system. Two air hoses must be connected to the rear top interfaces on the MR system.

Supply of air to the examination room will be covered by supply of air between ceiling and false ceiling and partly by supply of diffusers, mounted in the false ceiling. Refer to the Planning Reference Book for details.

11.2. ADJUSTMENT (BY COOLING SYSTEM SUPPLIER)

During the MR installation, before scanning the cooling system must be adjusted. This adjustment must be carried out by local air-conditioning personnel.

An often used method is to measure the average air speed V [m/s] in the tubes (diam 180 mm) and calculate the corresponding air-flow. To achieve the average air speed, several points in the pipe must be measured.

The air flow can be calculated as follows:

- Surface area A [m²] of each pipe = $(1/4 \pi d^2)$ [m²].
- Q [m³/h] = A [m²] $\times$ V [m/s] $\times$ 3600 [s]
- Each pipe is 180 mm in diameter so:
- Q [m³/h] = V [m/s] $\times$ 91.6

For 800 m³/hour (through each pipe 400 m³/h) the average air speed through each pipe can be calculated as:

- V [average] = 400 [m³/h] / 91.6
- V [average] = 4.36 m/s

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

12. COOLING SYSTEM POWER AND MASTER GRADIENTS

12.1. INTRODUCTION

Due to the fact that the gradient coil is water cooled, the air cooling system must be adjusted to a lower value.

The water cooled gradient coil must be cooled by an air-flow of 500 - 600 m³/h. The air is derived from the top through air diffusers in the top covers..

12.2. ADJUSTMENT (BY COOLING SYSTEM SUPPLIER)

Before scanning the cooling system must be re-adjusted. This adjustment must be carried out by local air-conditioning personnel.

An often used method is to measure the average air speed V [m/s] in the tubes (diam 180 mm) and calculate the corresponding air-flow. To obtain the average air speed, several points in the pipe must be measured.

The air flow can be calculated as follows:

- Surface area A [m²] of each pipe = $(1/4 \pi d^2)$ [m²]
- Q [m³/h] = A [m²] x V [m/s] x 3600 [s]
- Each pipe is 180 mm in diameter so:
- **Q [m³/h] = V [m/s] x 91.6**

For 500 m³/hour (through each pipe 250 m³/h) the average air speed through each pipe can be calculated as:

- V [average] = 250 [m³/h] / 91.6
- V [average] = 2.73 m/s through each hose.